

Data Cleaning & Essential Functions

Assignment



- What is data cleaning, and why is it important in data analysis?
 - What are the potential consequences of analyzing unclean or messy data?
 - Explain the common steps involved in cleaning and organizing data.
- How would you sort the following dataset first by "Department" (A-Z) and then by "Salary" (Largest to Smallest)? Write a step-by-step approach.

Employee	Department	Salary
Sonu	IT	4000
Pranav	HR	5000
Rahul	IT	2500

- Explain the use of text functions such as TRIM, LEFT, RIGHT, MID, and CONCAT in data cleaning.
- What is the role of date functions like TODAY in managing datasets?
- Apply Data Validation to restrict Quantity values to only whole numbers between 1 and 10.
 - Configure an input message that appears when a user selects a cell in the "Quantity" column explaining: "Please enter a whole number between 1 and 10."
 - Set up an error alert message that triggers if the user enters a number less than 1 or greater than 10, showing: "Invalid input! The quantity must be a whole number between 1 and 10."

Write a step-by-step approach for this question

Customer Name	Product Name	Category	Quantity	Unit Price (\$)
Jane Smith	Shoes	Electronics		81
Isabella Moore	Laptop	Electronics		121
Daniel Davis	Sofa	Clothing		239
Alex Moore	Shoes	Electronics		500
Michael Johnson	Table Lamp	Home Decor		423
Daniel Johnson	Backpack	Electronics		160
Isabella Davis	Headphones	Electronics		348
Jane Davis	Headphones	Electronics		152
Alex Wilson	T-shirt	Home Decor		369

- Understand and apply fundamental text functions like LEFT, RIGHT, MID, and LEN.
 - Extract the first 5 characters from the string "ExcelTipsAreGreat" using the LEFT function.
 - Extract the last 4 characters from "DataAnalysis.xlsx" using the RIGHT function.
 - Extract the substring "Tips" from "ExcelTipsAreGreat" using the MID function.
 - Count the total number of characters in the string "Hello World!" using the LEN function.
 - Create a formula to extract the middle 6 characters from "12345-67890-ABCDE".

7. Understand how to combine text using CONCAT, TEXTJOIN, and the & operator.
 - a. Use CONCAT to combine "Hello" and "World" with a space in between.
 - b. Combine "Apple", "Banana", and "Cherry" into a single string separated by commas using TEXTJOIN.
 - c. Use the & operator to create the string "2025: Excel Functions" by combining "2025", ":", and "Excel Functions".
 - d. Create a comma-separated list from the range A1:A5 using TEXTJOIN
 - e. Combine first names in column A with last names in column B to create full names in column C.
8. Understanding TODAY() and NOW()
 - a. What is the difference between TODAY() and NOW() in Excel? Provide an example of when you would use each function.
 - b. If cell A1 contains the date 2025-06-10, write a formula using TODAY() to determine how many days are left until that date
 - c. Write an Excel formula using NOW() to display the current date and time in the format MM/DD/YYYY HH:MM AM/PM.
 - d. If a cell contains =TODAY(), what will happen when the worksheet is reopened the next day? Explain
 - e. You want to store a static date (today's date) in a cell without it changing every day. What keyboard shortcut should you use?

